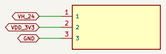
DIGITAL RESEARCH CIRCUIT / RESTRICTED 24 V, 3 mA SOURCES / ARTIFICIAL LOADS ONLY

Sixteen selectable stimulation contacts, two dedicated monitored returns; command-level routing is not measured continuity or isolation.

J1



QUALIFIED EXTERNAL RAILS

J2

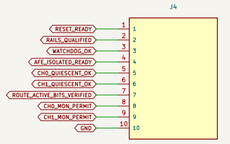


CANDIDATE PRELOAD AND COMMIT

J3

SOURCE ROUTE INTENT / PHASE STATUS

J4



EXTERNAL OBSERVED HEALTH CONTRACTS

GNDPWR



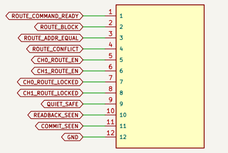
GNDPWR



GNDPWR

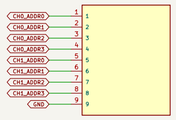


J5



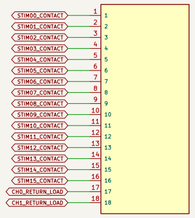
COMMAND STATUS / NOT MUX READBACK

J6



ACTIVE ADDRESS BIT READBACK

J7



SIXTEEN CONTACTS / TWO MONITORED RETURNS

Required before COMMIT: both WANT/REQUEST/RUN low, independently observed quiescence, monitor permits, rails, recording isolation and watchdog valid.

Then: COMMIT -> read all eight active Q bits -> fresh verification ACK edge -> separate REARM -> assert route WANT -> allow settling before phase.

source0

File: source0.kicad_sch

source1

File: source1.kicad_sch

address

File: address.kicad_sch

admission

File: admission.kicad_sch

collision

File: collision.kicad_sch

No mux state sensor, independent terminal cut-off, compliance detector, independent watchdog, rail generator, PCB or implanted device is completed here.

Native digital route slice; artificial loads; external protection/power/PCB remain

Project ATO

Sheet: /

File: contact_router.kicad_sch

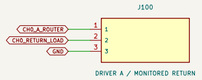
Title: ATO S4 / Route interfaces and monitored returns

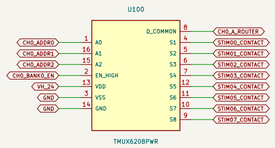
Size: A2	Date: 2026-10-13	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 1/6

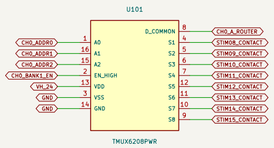
DIGITAL RESEARCH CIRCUIT / RESTRICTED 24 V, 3 mA SOURCES / ARTIFICIAL LOADS ONLY

Sixteen selectable stimulation contacts, two dedicated monitored returns; command-level routing is not measured continuity or isolation.

Both bank D commons share A_ROUTER. EN0 = ROUTE_EN & !ADDR3; EN1 = ROUTE_EN & ADDR3. Addresses 000..111 select S1..S8.

























Dedicated return: external contact -> RETURN_LOAD -> existing monitor 2 ohm shunt -> RETURN_DRIVER -> current-driver B_BR.

Do not bypass the monitor, add a second shunt, tie the two returns, or use recording REF/BIAS as a stimulation return.

TMUX6208 is rail-to-rail at valid 24 V supply; powered-off, stuck-on, parasitic coupling and two-source shared tissue paths remain unqualified.

A_ROUTER needs external terminal guard qualification before connection to A_BR. Internal enable-low is only a commanded off state.

Native digital route slice; artificial loads; external protection/power/PCB remain

Project ATO

Sheet: /source0/

File: source0.kicad_sch

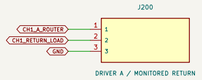
Title: ATO S4 / Source 0 / sixteen-contact output

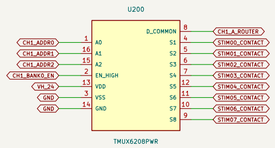
Size: A2	Date: 2026-10-13	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 2/6

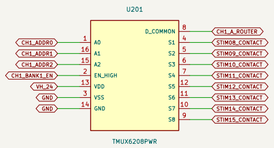
DIGITAL RESEARCH CIRCUIT / RESTRICTED 24 V, 3 mA SOURCES / ARTIFICIAL LOADS ONLY

Sixteen selectable stimulation contacts, two dedicated monitored returns; command-level routing is not measured continuity or isolation.

Both bank D commons share A_ROUTER. EN0 = ROUTE_EN & !ADDR3; EN1 = ROUTE_EN & ADDR3. Addresses 000..111 select S1..S8.











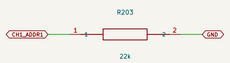














Dedicated return: external contact -> RETURN_LOAD -> existing monitor 2 ohm shunt -> RETURN_DRIVER -> current-driver B_BR.

Do not bypass the monitor, add a second shunt, tie the two returns, or use recording REF/BIAS as a stimulation return.

TMUX6208 is rail-to-rail at valid 24 V supply; powered-off, stuck-on, parasitic coupling and two-source shared tissue paths remain unqualified.

A_ROUTER needs external terminal guard qualification before connection to A_BR. Internal enable-low is only a commanded off state.

Native digital route slice; artificial loads; external protection/power/PCB remain

Project ATO

Sheet: /source1/

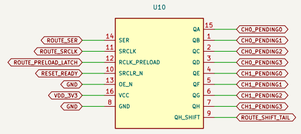
File: source1.kicad_sch

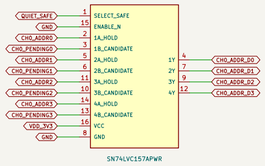
Title: ATO S4 / Source 1 / sixteen-contact output

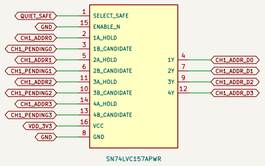
Size: A2	Date: 2026-10-13	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 3/6

DIGITAL RESEARCH CIRCUIT / RESTRICTED 24 V, 3 mA SOURCES / ARTIFICIAL LOADS ONLY

Sixteen selectable stimulation contacts, two dedicated monitored returns; command-level routing is not measured continuity or isolation.







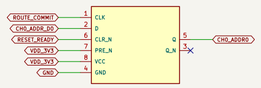






Candidate bits may change without changing active addresses. Eight active flip-flops always see raw COMMIT clock; safe D selection loads or holds.

RESET_READY low asynchronously clears active bits to zero; outputs remain inhibited until legal COMMIT, fresh readback ACK and deliberate REARM.



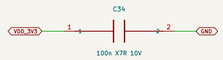


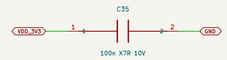


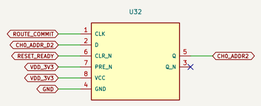




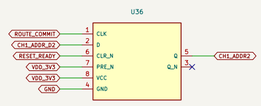


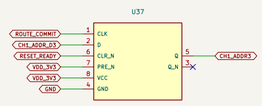
















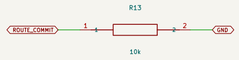














SRCLR affects only the candidate shift stage. QH_SHIFT is payload cascade, not active-address or analog switch readback. Actual Q bits appear at J6.

QUIET_SAFE and candidate data must meet physical setup/hold. Gate delay, asynchronous clear, metastability and clock-integrity qualification remain.

Native digital route slice; artificial loads; external protection/power/PCB remain

Project ATO

Sheet: /address/

File: address.kicad_sch

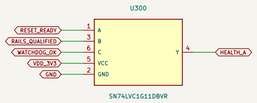
Title: ATO S4 / Candidate preload and active addresses

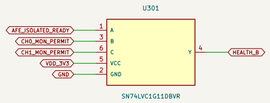
Size: A2	Date: 2026-10-13	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 4/6

DIGITAL RESEARCH CIRCUIT / RESTRICTED 24 V, 3 mA SOURCES / ARTIFICIAL LOADS ONLY

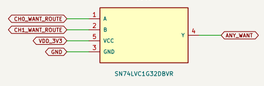
Sixteen selectable stimulation contacts, two dedicated monitored returns; command-level routing is not measured continuity or isolation.

BASE_HEALTH = RESET & RAILS & WATCHDOG & AFE_ISOLATED_READY & MONO & MON1; QUIET_SAFE additionally requires both WANT/REQUEST/RUN low and quiescence.



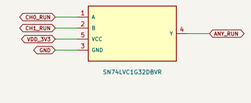


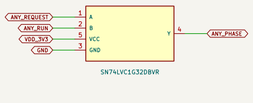


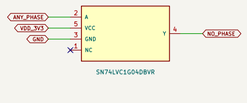


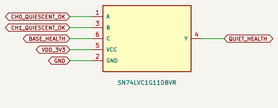


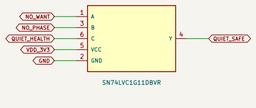




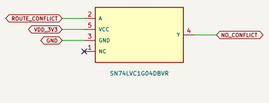


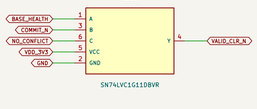


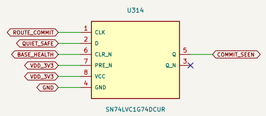


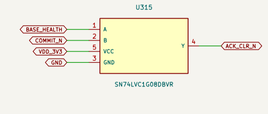


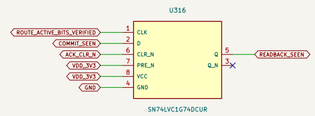


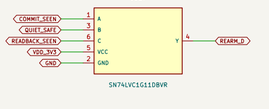


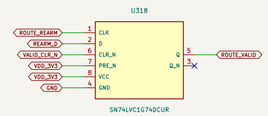


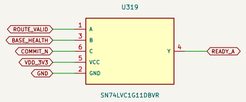


























































































Any COMMIT invalidates prior validation and ACK. Illegal COMMIT holds active bits and captures COMMIT_SEEN=0.

Rearm needs a fresh post-commit ACK edge. Health recovery or collision removal cannot create a clock edge or restart output.

Ideal logic only: setup/hold, all-off settling, voltage, independent watchdog and actual current observation remain mandatory.

Native digital route slice; artificial loads; external protection/power/PCB remain

Project ATO

Sheet: /admission/
File: admission.kicad_sch

Title: ATO S4 / Commit, acknowledgement and rearm

Size: A2	Date: 2026-10-13	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 5/6

DIGITAL RESEARCH CIRCUIT / RESTRICTED 24 V, 3 mA SOURCES / ARTIFICIAL LOADS ONLY

Sixteen selectable stimulation contacts, two dedicated monitored returns; command-level routing is not measured continuity or isolation.

Conflict = WANT0 & WANT1 & equal(all four active address bits). It disables BOTH bank families and asynchronously clears route validation.







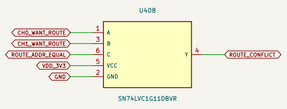


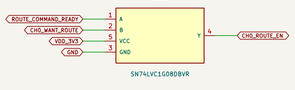




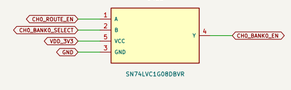




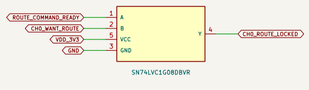


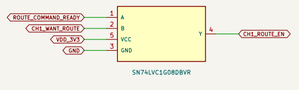








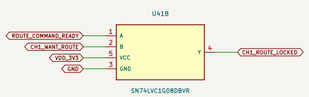










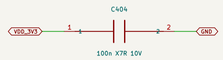




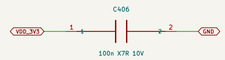
















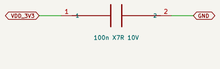


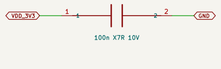


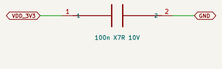


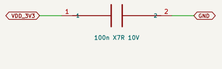


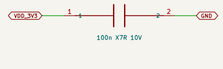












Removing a conflict only removes its fault condition; it does not restore VALID. Release both WANTs, verify quiet, then explicitly rearm.

The two source enables may coexist only at different contacts. Common tissue impedance and volume conduction are outside this digital collision test.

ROUTE_LOCKED means admitted digital command ownership only. No selected-contact continuity or power-off isolation is inferred.

Native digital route slice; artificial loads; external protection/power/PCB remain

Project ATO

Sheet: /collision/
File: collision.kicad_sch

Title: ATO S4 / Conflict and bank enables

Size: A2	Date: 2026-10-13	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 6/6